

Evaluation of different methods to estimate incoming solar radiation

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ABSTRACT

Unavailability of daily solar radiation data leads to use alternate ways for estimation of this most important weather variable. The present study was carried out to evaluate the methods of estimating incoming solar radiation using two alternate data sources i.e. ground observatory data (sunshine hours based Ångström-Prescott model; temperature based Bristow–Campbell model and Hargreaves model and both temperature and cloud amount based Supit model) and satellite based Earth Radiation Budget Experiment (ERBE) data.

Ångström-Prescott model was found to be the best among different empirical models tried, with highest R^2 (0.72) and lowest RMSE (2.86 MJ m^{-2}) followed by Supit ($R^2=0.44$ and $\text{RMSE}=3.97 \text{ MJ m}^{-2}$). Two methods based only on temperature viz., Hargreaves and Bristow-Cambell remained at par with R^2 value of 0.36 and 0.34, respectively. Validation was done using the station-wise empirical coefficients and found that all the models viz. temperature based Hargreaves' and Bristow-Cambell models, the temperature and cloud amount based Supit's model resulted with similar R^2 (0.64, 0.65 and 0.66 respectively) and RMSE (2.86 , 2.82 and 2.80 MJ m^{-2} , respectively). Satellite derived solar radiation showed R^2 of 0.63 with lowest RMSE of 2.08 MJ m^{-2} .

Key words: Solar radiation, ERBE, Ångström-Prescott model, Bristow–Campbell model, Hargreaves model, Supit mode

The increasing interest among agricultural researchers in using crop growth simulation models as tools for spatial yield forecasting, has led to an urgent requirement of detailed spatial daily weather data like global solar radiation maximum and minimum air temperatures, and rainfall. Globally, the number of ground-level weather stations recording solar radiation is limited compared to the number of stations recording air temperature and precipitation (Richardson, 1985; McVicar and Jupp, 1999; Thornton and Running, 1999; Liu and Scott, 2001; Mandal et al., 2003). Early contribution to deal with the problem lead to the recommendation for the use of average values, spatial interpolation, estimation of global radiation values using remote sensing data or estimation of these values from other climatic variables. However, due to the variation in daily and annual global radiation and the nonlinear relation between radiation and photosynthesis, the use of average data (even over short periods) to replace missing data must be avoided in crop growth simulation model for yield prediction (Nonhebel, 1994). Spatial interpolation is not always an option since the density of recording stations is too

sparse in India (only 18-22 regular reporting among 45 solar radiation recording stations). Hence, the alternate ways for getting accurate spatial solar radiation data are either from satellite data (Gupta et al., 2001; Ba and Nicholson, 2001; Kimothi et al., 2004) or from other weather variables such as sunshine duration (Ångström, 1924; Boisvert *et al.*, 1990); air temperature range (Bristow & Campbell, 1984; Hargreaves *et al.*, 1985; De Jong & Stewart, 1993;), precipitation (De Jong & Stewart, 1993) and cloud-cover (Barker, 1992; Supit, 1994) using empirical equation. There is a need to evaluate the existing methods based on availability of data and its simplicity to apply over a region. Recently, Mandal et al.(2003) evaluated nine empirical approaches for estimating solar radiation at Hyderabad in India and suggested similar studies for other stations.

The present study was carried out with the objective to validate the surface solar radiation estimated using satellite data and evaluate its accuracy with methods of incoming solar radiation estimation using ground observatory data (sunshine hours, maximum and minimum temperature, cloud amount).

MATERIALS AND METHODS

Estimation of solar radiation from satellite data

The data of daily surface solar radiation estimates for a period of 10 years (1983-1993) at $1^\circ \times 1^\circ$ grid over India ($4^\circ\text{N} - 36^\circ\text{N}$ and $67^\circ\text{E} - 98^\circ\text{E}$) were obtained from National Aeronautics and Space Administration (NASA) Langley Research Center Distributed Active Archive Center (DAAC) (Gupta et al. 1999). The daily average global surface solar radiation was derived with quality-check sw (QCSW) algorithm of NASA World Climate Research Programme/Global Energy and Water-Cycle Experiment (WCRP/GEWEX) Surface Radiation Budget (SRB) Project (Gupta et al. 2001). The algorithm uses satellite and operational meteorological data for inputs. The daily ground observed solar radiation data obtained from International Crops Research Institute for the Semi Arid Tropics (ICRISAT), Patancheru, Hyderabad (17.45°N latitude and 78.47°E longitude) from July 1, 1983 to June 30, 1993 (10 years) were used to validate the satellite data of the corresponding grid (Fig. 1).

Estimation of solar radiation using ground observatory data

Daily data of solar radiation, bright sunshine hour, maximum temperature, minimum temperature, precipitation and cloud cover for 19 different weather stations collected by Indian Meteorological Department for the period 1997 to 2000 were used. The details about the latitude, longitude and altitude (elevation) for the stations are given in Table 1. The stations selected represent the different agroclimatic regions of India. Three years data (1997-1999) were used to fit the empirical equation and determine the regression coefficients. One-year data of the year 2000 were used for validation of the derived equation.

Four different empirical equations developed earlier were used to compute solar radiation and were compared for their applicability in different region on the basis of the availability of weather data and accuracy. The detail of each equation is given below.

Ångström and Prescott's model

Ångström (1924) developed a model for estimating the solar radiation reaching the earth's

surface by using the relationship of the measured sunshine duration and maximum day length with total radiation under a real atmosphere for completely clear days. Prescott (1940) introduced an improved version of the original formula where R_s is based on the fraction of actual and potential sunshine duration during the day:

$$R_s = R_a \left(A_a + B_a \frac{n}{N} \right) \quad \dots\dots\dots(1)$$

Where, R_s is Incoming global solar radiation ($\text{MJ m}^{-2} \text{d}^{-1}$), R_a is extra-terrestrial radiation ($\text{MJ m}^{-2} \text{d}^{-1}$) (eq 6), A_a and B_a are empirical constants, n is observed bright sunshine hours per day (hr) and N is astronomical day length (hr). The regression constants A_a and B_a have a physical meaning: A_a can be considered as the fraction of extra terrestrial radiation on overcast days. Sum of A_a and B_a can be considered as the fraction of radiation received on clear days. The Ångström-Prescott method is easy to apply when data for bright sunshine hours are available.

Bristow – Campbell Model

Daily maximum temperature decreases with reduced transmissivity (i.e. increased cloud cover), while minimum air temperature will increase due to the cloud emissivity. Conversely, clear skies will increase maximum air temperature due to higher short wave radiation input, and minimum air temperature will decrease due to higher transmissivity. Based on the principle, Bristow and Campbell (1984) developed a model to compute the solar radiation reaching the ground (R_s) as the fraction of daily total atmospheric transmittance of extra terrestrial radiation (R_a).

$$R_s = R_a A_b \left\{ 1 - \exp(-B_b \Delta T^{C_b}) \right\} \quad \dots\dots\dots(2)$$

Where, the term $A_b \left\{ 1 - \exp(-B_b \Delta T^{C_b}) \right\}$ is the atmospheric transmissivity (T_i) as a function of temperature range (ΔT). The ΔT can be computed as:

$$\Delta T = T_{\max}(I) - \frac{[T_{\min}(I) + T_{\min}(I+1)]}{2}$$

Table 1: The Selected Station and their details

Sr. No.	Station name	Latitude (deg N)	Longitude (deg E)	Altitude (m)
1.	Ahmedabad	23.07	72.63	55
2.	Bangalore	12.97	77.58	921
3.	Chennai	13.00	80.18	16
4.	Dehradun	30.32	78.03	682
5.	Jaipur	26.82	75.80	390
6.	Jaisalmer	26.90	70.92	231
7.	Jodhpur	26.30	73.02	224
8.	Kolkata	22.65	88.45	6
9.	Minicoy	8.30	73.15	2
10.	Mumbai	19.12	72.85	14
11.	New Delhi	28.58	77.20	216
12.	Okha	22.48	69.12	7
13.	Panjim	15.48	73.82	60
14.	Patna	25.60	85.17	57
15.	PortBlair	11.67	92.72	79
16.	Pune	18.53	73.85	559
17.	Ranchi	23.32	85.32	652
18.	Srinagar	34.08	74.83	1587
19.	Thiruvananthapuram	8.48	76.95	64
20.	Hyderabad	17.45	78.47	545

Where, $T_{\max}(I)$ is maximum temperature ($^{\circ}\text{C}$) of I^{th} day and $T_{\min}(I)$ and $T_{\min}(I+1)$ are minimum temperature ($^{\circ}\text{C}$) of the same day and next day, respectively.

The empirical constant A_b represents maximum clear sky characteristic of the area, which depends on the elevation of the place and pollution contents of the air. B and C are functions of time and determine how soon maximum T_i is achieved as ΔT increases.

Hargreaves Model

It is a relatively simple method of estimating solar radiation by relating the daily temperature range (difference between maximum and minimum temperature to global radiation (Hargreaves et al., 1985).

$$R_s = R_a A_h \sqrt{(T_{\max} - T_{\min})} + B_h \dots\dots\dots(3)$$

Where, A_h and B_h are the empirical constants for Hargreaves formula.

Supit Model

When observations for sunshine duration are not available, the equation that uses cloud cover observations and the temperature range as input can be applied (Supit, 1994; (Supit and van Kappel, 1998). Clouds and their accompanying weather patterns are among the most important atmospheric phenomena restricting the availability of solar radiation at the earth surface. Various studies to estimate global solar radiation from observations of various cloud layer amount and cloud types have been executed (e.g. Davies & McKay, 1988; Barker, 1992).

$$R_s = R_a \left(A_s \sqrt{(T_{\max} - T_{\min})} + B_s \sqrt{\left(1 - \frac{CC}{8}\right)} \right) + C_s \dots\dots(4)$$

Where, CC is the cloud cover in oktas and A_s , B_s and C_s are the empirical constants for Supit formula.

The models based on precipitation data (McCaskill, 1990), both temperature and precipitation (De Jong and Stewart, 1993, Hunt et al., 1998, Liu and

Scott, 2001) were not tested because Mandal *et al.* (2003) tried for Hyderabad in India and showed that they gave similar results as the models with temperature alone.

Estimation of Extra terrestrial radiation (R_a)

The amount of radiation received at the top of the atmosphere, i.e. extra terrestrial radiation (R_a) is dependent on latitude and the time of year only if radiation from the sun perpendicular to the sun's rays assumed to be constant. Because the earth's axis is tilted and the distance from the sun to earth varies with season, the geometry involved is complex but merely represents an integration of the functional relationship between zenith angle of sun with an empirical distance correction.

$$R_a = SC \left[\frac{\bar{d}}{d} \right]^2 \int_{Sunset}^{Sunrise} \cos \theta_z dt \quad \dots\dots\dots(5)$$

Where, SC is solar constant, its value is 118.1 MJ m⁻² day⁻¹, $\frac{\bar{d}}{d}$ is Duffie-Beckman sun-earth distance correction in which d is the distance of earth at any time and $\bar{d}$ is the average distance between the sun and the earth.

$$\frac{\bar{d}}{d} = 1 + 0.034 \cos \left[\frac{360D}{365.25} \right] \quad \dots\dots\dots \text{Duffie and Beckman (1980).}$$

θ_z , is zenith angle of the sun. Sellers(1965) and Garg(1982) computed $\cos \theta_z$ as:

$$\cos \theta_z = \sin L \sin \delta + \cos L \cos \delta \cos \omega_h$$

Where, L is the latitude of the place in degrees, δ is the declination of the sun and ω_h is an hour angle. A procedure developed by Jacquot and Porter (1989) for the Fourier analysis of periodic functions was used to develop a cosine series to predict δ . Three terms of the series were evaluated. The use of two terms shown below is recommended because the algorithm is only slightly more complicated than that of Cooper (Cooper, 1969).

$$\delta = 0.3931 + 23.2577 \cos(0.9861D - 170.7) + 0.3906 \cos(1.9154 - 174.4)$$

A general expression for the hour angle ω_h is:

$$\cos \omega_h = \sin [\cos \theta_z / \cos L \cos \delta] - \tan \delta \tan L$$

Because the $\cos \theta_z$ is zero at sunset or sunrise, the hour angle at sunrise and sunset reduces to the following:

$$\cos \omega_{hs} = -\tan \delta \tan L \quad ; \text{ where, } \omega_{hs} \text{ is the sunset hour angle.}$$

Sellers (1965) gave the detailed derivation of the above equation as:

$$R_a = \frac{SC}{\pi} \left[\frac{\bar{d}}{d} \right]^2 \left[\omega_{hs} \left\{ \frac{\pi}{180} \right\} \sin L \sin \delta + \cos L \cos \delta \sin \omega_{hs} \right] \quad \dots\dots\dots(6)$$

Evaluation Statistics

The performance of each model was assessed on the basis of coefficient of determination (R^2) and root mean square error (RMSE), which give an idea of the magnitude of the non-systematic error. The RMSE calculated in actual units i.e., MJ m⁻² as:

$$RMSE = \sqrt{\frac{\sum (R_{obs} - R_{est})^2}{N}} \quad \dots\dots\dots(7)$$

Where, R_{obs} and R_{est} are observed and estimated solar radiation and N is the total number of observations.

RESULTS AND DISCUSSION

Estimation of Coefficients using different models

The model coefficients were computed for four different methods (Ångström, Hargreaves, Supit and Bristow – Campbell) using three years data for 19 stations (1997-1999) and Hyderabad (July-1990 to June-1993) and presented in Table 2. Coefficients were also generated for *rabi* (November to April) and *kharif* (June to October) seasons separately. The R^2 and RMSE values that shows the strength and error of the best fit, were computed for all the stations using four methods as shown in Table 2.

Table 2: Coefficients generated using different empirical methods for estimation of solar radiation for different stations in India

Sr. No.	Station name	Angstrom-Prescott Method				Supit Method				Hargreave's Formula				Bristow Campbell Model					
		A _s	B _s	R ²	RMSE	A _s	B _s	C _s	R ²	RMSE	A _h	B _h	R ²	RMSE	A _b	B _b	C _b	R ²	RMSE
1	Ahmedabad	0.26	0.42	0.84	2.97	0.156	-0.008	-0.179	0.63	2.76	0.157	-0.374	0.63	2.75	0.69	0.125	1.034	0.58	2.51
2	Bangalore	0.27	0.51	0.74	2.85	0.005	0.207	-5.300	0.44	4.19	0.189	-3.196	0.38	4.41	0.74	0.081	1.163	0.31	4.65
3	Chennai	0.25	0.48	0.84	2.84	0.150	0.371	-1.536	0.41	4.02	0.157	1.206	0.31	4.34	0.62	0.125	1.206	0.23	4.28
4	Dehradun	***	***	***	**	0.120	0.224	2.877	0.60	2.18	0.157	0.854	0.55	2.31	0.71	0.105	1.137	0.67	2.06
5	Jaipur	***	***	***	**	0.146	0.121	2.369	0.54	3.69	0.147	3.012	0.53	3.70	0.78	0.279	0.683	0.30	3.45
6	Jaisalmer	0.49	0.17	0.12	2.53	0.147	0.025	2.790	0.72	2.65	0.147	2.837	0.72	2.60	0.66	0.128	1.275	0.06	2.59
7	Jodhpur	0.33	0.33	0.38	2.24	0.151	0.131	-0.013	0.64	2.88	0.152	0.378	0.62	2.94	0.63	0.050	1.561	0.29	2.36
8	Kolkata	***	***	***	**	0.119	0.214	2.548	0.30	3.97	0.129	2.978	0.24	4.15	0.65	0.196	0.888	0.24	3.96
9	Minicoy	0.33	0.38	0.81	1.99	-0.005	1.524	0.066	0.99	0.33	0.207	-0.624	0.12	5.42	0.52	0.001	5.019	0.33	4.72
10	Mumbai	0.31	0.41	0.76	2.89	0.186	0.413	-2.919	0.39	4.14	0.136	5.014	0.17	4.82	0.63	0.186	1.204	0.28	4.23
11	New Delhi	0.34	0.42	0.70	2.77	0.153	0.160	-0.669	0.59	3.77	0.154	-0.110	0.58	3.74	0.64	0.051	1.500	0.23	4.02
12	Okha	0.41	0.33	0.63	2.22	0.211	-0.108	6.232	0.60	2.51	0.177	7.536	0.53	2.73	0.72	0.885	0.553	0.20	3.04
13	Panjim	0.27	0.41	0.90	1.66	0.216	0.519	-7.277	0.45	3.30	0.149	3.635	0.16	4.08	0.64	0.043	2.000	0.58	2.61
14	Patna	0.24	0.41	0.86	1.77	0.119	0.309	1.850	0.43	3.51	0.096	6.488	0.29	3.94	0.65	0.384	0.580	0.22	3.71
15	PortBlair	***	***	***	**	0.016	1.357	-0.560	0.89	1.75	0.147	4.728	0.14	4.78	0.63	0.076	1.500	0.07	5.20
16	Pune	0.30	0.45	0.86	2.75	0.129	0.281	0.940	0.61	2.94	0.115	4.507	0.49	3.33	0.70	0.258	0.716	0.49	3.07
17	Ranchi	0.24	0.53	0.55	2.00	0.148	-0.139	4.421	0.60	2.64	0.144	4.378	0.57	2.73	0.80	0.077	1.224	0.43	2.60
18	Srinagar	0.37	0.16	0.08	4.10	0.019	0.987	5.479	0.70	3.25	0.111	1.700	0.47	4.33	0.78	0.146	0.522	0.14	3.92
19	Thiruvananthapuram	0.37	0.34	0.75	1.76	0.028	1.437	-2.024	0.98	0.58	0.331	-12.973	0.51	3.00	0.65	0.052	1.779	0.5	3.06
20	Hyderabad	0.29	0.39	0.81	1.95	***	***	***	***	**	0.161	-0.492	0.63	2.74	0.64	0.044	1.593	0.71	2.38
	Pooled	0.30	0.42	0.72	2.86	0.138	0.125	1.778	0.44	3.97	0.127	3.976	0.36	4.23	0.70	0.180	0.923	0.34	3.53

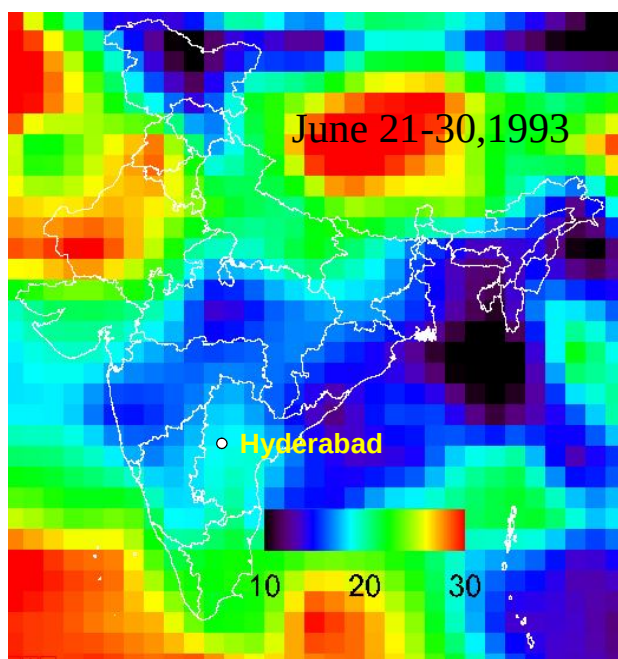


Fig. 1: An example of Satellite derived solar radiation (MJ m^{-2}) over India (Data source: NASA LaRC DAAC, averaged over June 21-30, 1993).

Evaluation of different methods

The Ångström's coefficients were estimated for 16 stations out of 20 selected stations because the data of bright sunshine duration were not available for 4 stations (Table 2). Based on the best fit among the four methods, the Ångström formula (based on sunshine duration) showed highest R^2 (0.72) and lowest RMSE (2.86 MJ m^{-2}) followed by the Supit model (based on temperature and cloud amount) with R^2 and RMSE of 0.44 and 3.97 MJ m^{-2} , respectively. Two methods based only on temperature viz., Hargreaves and Bristow-Cambell remained at par with R^2 0.36 and 0.34, respectively (Table 2). Individually, 12 stations out of 16 showed the best fittings with R^2 greater than 0.7 for Ångström method. Two stations, Jaisalmer and Srinagar, with very poor fittings for Ångström (R^2 about 0.1), showed better fittings for Supit with R^2 of 0.72 and 0.48 and for Hargreaves with R^2 of 0.72 and 0.47, respectively. The better performance of the Supit method by about 20-25 per cent over Hargreaves

and Bristow-Cambell shows the importance of cloud amount data in addition to temperature when bright sun shine hour data not available. For the stations situated on islands (Portblair and Minicoy) and peninsula (Mumbai and Panjim), effect of cloudy conditions throughout the year results into diffused radiation with different intensities having smaller value of daily temperature range (ΔT , i.e. less variable) and higher value of cloud amount (overcast, 8 octas). The effect is depicted from Table 2 with the poor performance of linear models based on temperature (Hargreaves method) and temperature and cloud amount (Supit model).

It was found that except few stations, the performance of the Bristow – Cambell model was poor compared to Hargreaves model for individual stations but it can be used as alternate method when radiation has the non-linear behaviour with temperature (Table 2 for Dehradun, Panjim and Hyderabad). It was also found that the strength of formula improved when it was developed separately for *rabi* and *kharif* season.

Validation of different methods based on radiation estimates

Equation developed using three years data (1997-99) were used to estimate the solar radiation for the year 2000. Due to non-availability of bright sunshine hour data for certain weather stations, Ångström equation was not tried for validation though it was found the best among four methods tried (Table 2, Supit, 1998). The validation results using station-wise separate coefficients are presented in Fig. 2. All the models viz. temperature based Hargreaves' and Bristow-Cambell models, the temperature and cloud amount based Supit's model resulted with similar R^2 (0.64, 0.65 and 0.66 respectively) and RMSE (2.86 , 2.82 and 2.80 MJ m^{-2} , respectively). Solar radiation computed with the use of different coefficients for each of the selected 20 stations gave better R^2 than coefficients generated for pooled data. Therefore, it is suggested to use separate coefficients for each station to estimate daily solar radiation. The satellite derived surface radiation estimates (1983-93) were validated with the ground observed radiation at ICRISAT, Hyderabad. The results indicated that satellite derived solar radiation showed

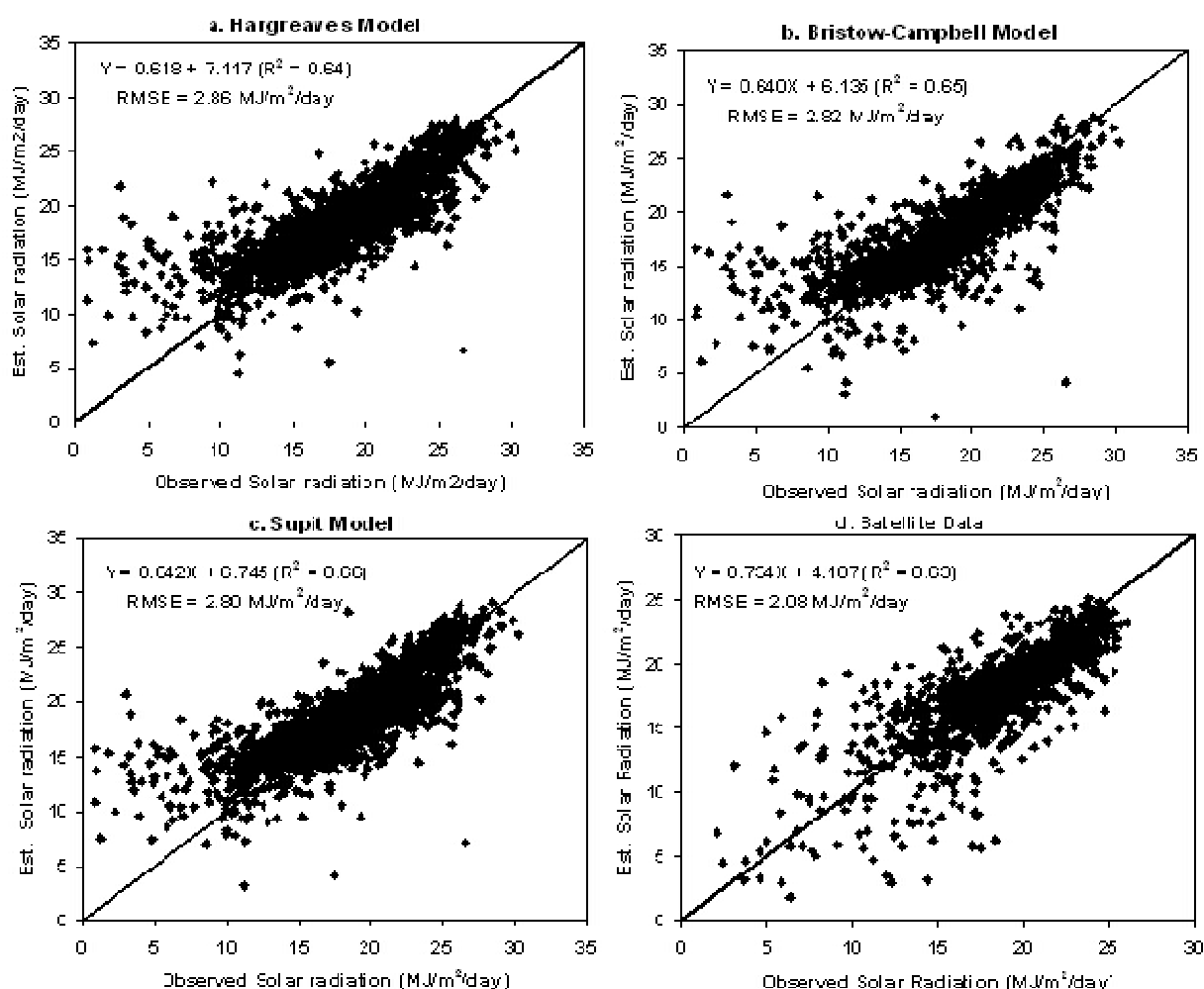


Fig. 2: Observed vs estimated solar radiation using different methods

the lowest RMSE of 2.08 MJ/m² as compared to the radiation estimated from various empirical models (Fig. 2) and R^2 comparable with the Hargreaves model (0.63).

CONCLUSIONS

Ångström-Prescott method is found to be the best, which uses sunshine hour as input parameter. The satellite derived radiation estimates showed promising results with the lowest RMSE of 2.08 MJ/m² and it needs to be validated at more number of stations over

India. However, in absence of satellite estimates and sunshine hour data temperature based Bristow-Campbell or Hargreaves model may be used. As these two methods resulted in more or less similar RMSE and comparable R^2 values so the simplest among two i.e. Hargreaves method is more practicable for estimating solar radiation. From the analysis it is it may be interpreted that it is better to use station-wise coefficients to estimate daily solar radiation than using the coefficients developed from the pooled data. The results are promising and the climatic set of coefficients for each agroclimatic zone may be generated using at least ten years data to improve the accuracy of estimates.

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